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Capacity Assignment in Diffserv Networks for Proportional QoS

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Abstract

The Differentiated Services (Diffserv) architecture is a scalable solution for providing Quality of Service (QoS) over packet switched networks. Yet, Diffserv cannot offer strict performance guarantees to its subscribers. We consider in this paper an alternative notion of *relative* performance guarantees. Specifically, the network manager's goal is to maintain pre-defined ratios between certain congestion measures over the different service classes. For that purpose, the manager may judiciously allocate available capacity between the service classes. This scheme is studied within a flow model, with self-optimizing users, which are subject to static SLAs. We pose the entire problem as a *non-cooperative game*. Starting with a simplified network model, we demonstrate the existence and uniqueness of the equilibrium, and show how to calculate the required capacity assignment. We then extend these results to a general network topology. Finally, the incorporation of dynamic SLAs is briefly addressed.